

Fabric Defect Detection and Localisation Using Deep Neural Networks

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1. ABSTRACT

Manual fabric inspection is slow and inconsistent, so automated defect detection is valuable for textile quality control. Using the AITEX dataset (247 images, 141 defect-free / 106 defective across 9 defect types), we built a deep-learning classifier combining CNN features (ResNet, MobileNetV3) with texture and patch-based features. This model reached 89.2% cross-validated accuracy and 0.94 AUC, far outperforming a classical texture-based baseline (61.5% accuracy). We compared it against two other approaches: an autoencoder that flags defects without needing labelled examples, and a U-Net that localises exactly where defects occur in an image. Together, these results highlight the strengths and trade-offs of supervised, unsupervised, and localisation-based approaches to defect detection.

2. INTRODUCTION

In textile manufacturing, fabric quality is typically verified through visual inspection, a human inspector examines rolls of fabric under good lighting, looking for flaws before the material is cut, dyed, or shipped. This process is critical, since a single undetected defect can render an entire batch unsellable, but it is also slow, subjective, and error-prone. Inspectors fatigue over long shifts, judge borderline cases inconsistently, and simply cannot keep pace with the speed of modern production lines. As demand for higher throughput grows, there is a clear need for automated systems that can inspect fabric faster and more reliably than manual review.

Building such a system, however, is a genuinely difficult computer vision problem. Fabric defects are rare relative to the amount of defect-free material produced, so there are few examples to learn from. They are also visually subtle and highly diverse: a hole, a stain, broken yarn, and a knot can look completely different from one another, meaning a model cannot rely on a single, consistent visual signature. Compounding this, the fabric itself has a repetitive, textured background pattern that can easily be confused with a genuine defect, especially at the pixel level. Together, these factors: rarity, diversity, and background noise make fabric defect detection a challenging and useful problem for testing a range of classical and deep learning techniques.

3. PROBLEM

The project uses the AITEX Fabric Image Database, a benchmark dataset for fabric defect detection. It contains 247 high-resolution images (4096 × 256 pixels) from seven fabric types, including 141 defect-free and 106 defective images. The defects belong to nine categories, such as holes, stains, broken yarn, and knots. Each defective image is provided with a binary mask that accurately marks the defect location, enabling evaluation of both defect detection and localization.

The dataset supports two related but distinct computer vision tasks: defect classification and defect localization. The classification task aims to determine whether a fabric image is defective or defect-free, while the localization task identifies the exact region of the defect within the image. Since a model that performs well in classification may not necessarily localize defects accurately, the two tasks must be evaluated independently using appropriate performance metrics. For Task 2, the AITEX test set is unlabelled (Kaggle-style). Therefore, all reported performance metrics are obtained using group-safe cross-validation on the training data rather than the held-out test set.

Despite its advantages, the dataset presents several challenges. Its small size (247 images) increases the risk of overfitting, while the class imbalance between defect-free and defective images can bias models toward the majority class. To increase the amount of training data, the large images are often divided into smaller patches and augmented. However, this must be done carefully to prevent data leakage, where patches from the same original image appear in both the training and validation sets. To ensure reliable evaluation, image-level splitting and group-safe cross-validation are essential.

7. REFERENCES

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4. METHODOLOGY

Task 1: Classical Baseline

Hand-crafted texture descriptors (LBP, GLCM, HOG, Gabor filters) extracted per image patch, with SMOTE oversampling for class balance. Compared SVM, Random Forest, and XGBoost classifiers, combined into a voting ensemble.

Task 2: Deep Feature Ensemble (CNN)

Each 4096×256 fabric image split into 16 non-overlapping 256×256 patches. Three frozen, ImageNet-pretrained CNNs (ResNet-18, ResNet-34, MobileNetV3) extract deep feature embeddings per patch, combined with GLCM texture statistics and patch-similarity metrics (~9,960 features, reduced to top 300 via ANOVA F-score). A 6-model ensemble (ExtraTrees, SVM, GradientBoosting, Logistic Regression, MLP, Random Forest) trained with soft voting, validated with Stratified 5-fold cross-validation.

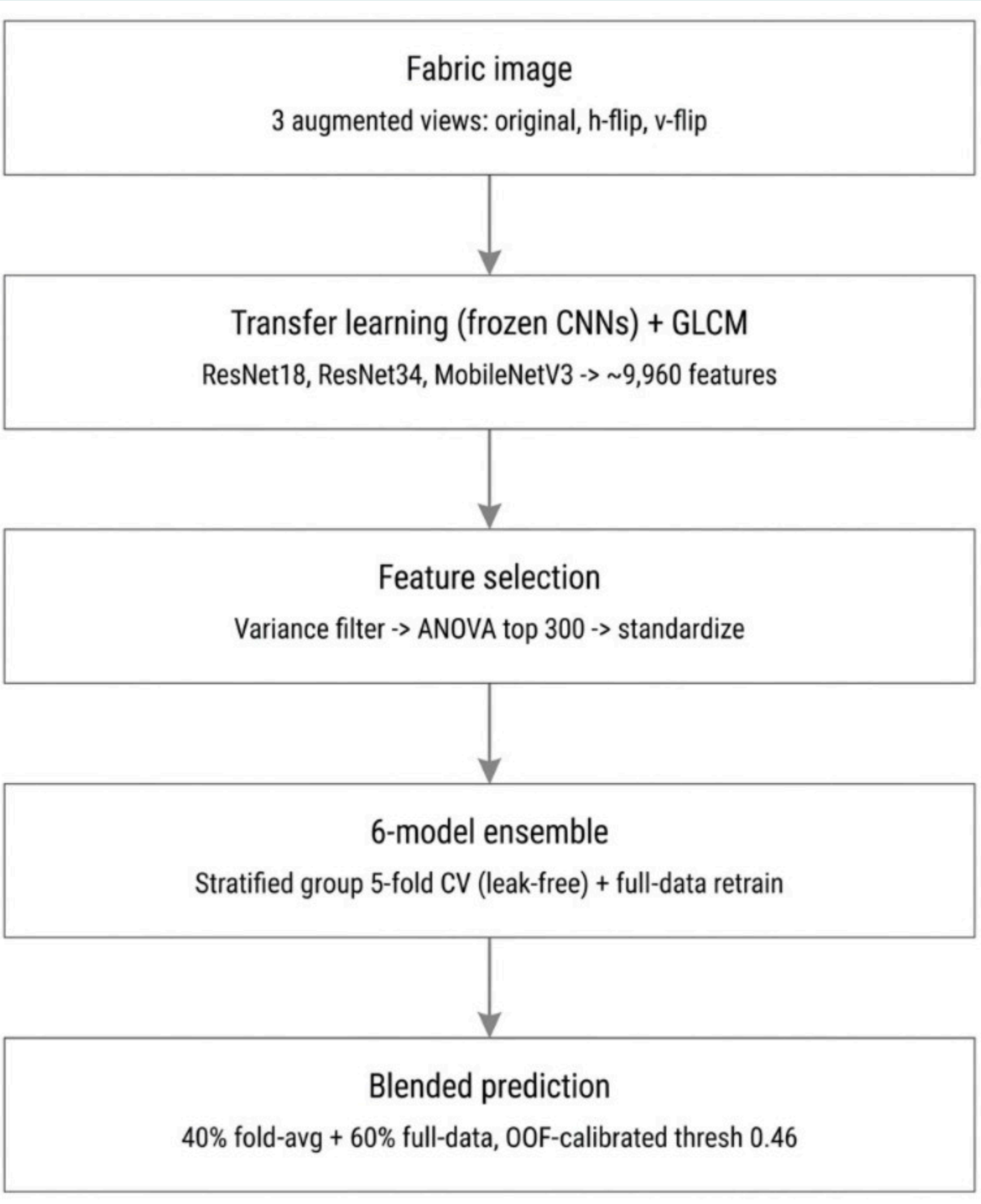
Task 3: Autoencoder (Anomaly Detection)

Deep Convolutional Autoencoder with Residual Blocks (6.33M params) trained strictly on defect-free 128×128 fabric patches. Reconstruction error scored at inference time using multi-scale (128px and 64px) dense sliding-window patches with Test-Time Augmentation (TTA), combined with an MSE + SSIM + Edge Loss training schedule. Extracted a 28-dimensional spatial anomaly feature vector per image and classified via a Gradient Boosting Classifier, with decision thresholds optimized using Youden's J statistic on held-out out-of-fold (OOF) calibration data.

Task 4: U-Net Segmentation

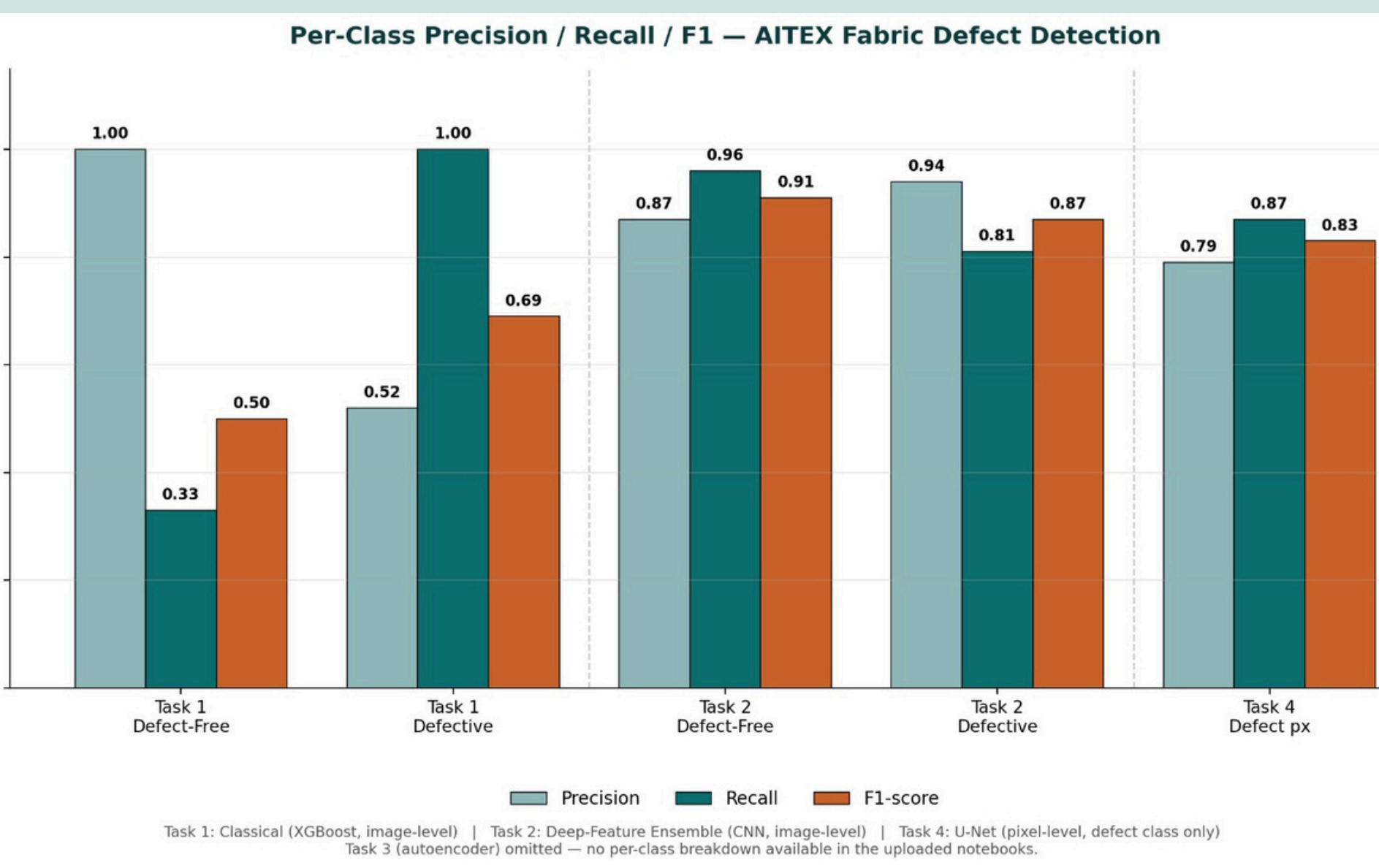
Defect images were paired with their ground-truth masks, then split into 128×128 patches for training. A standard U-Net (DoubleConv blocks, ~7.8M parameters) was trained on a stratified train/validation split using a combined Dice + BCE loss with the Adam optimizer and early stopping.

Final Workflow of Task 2

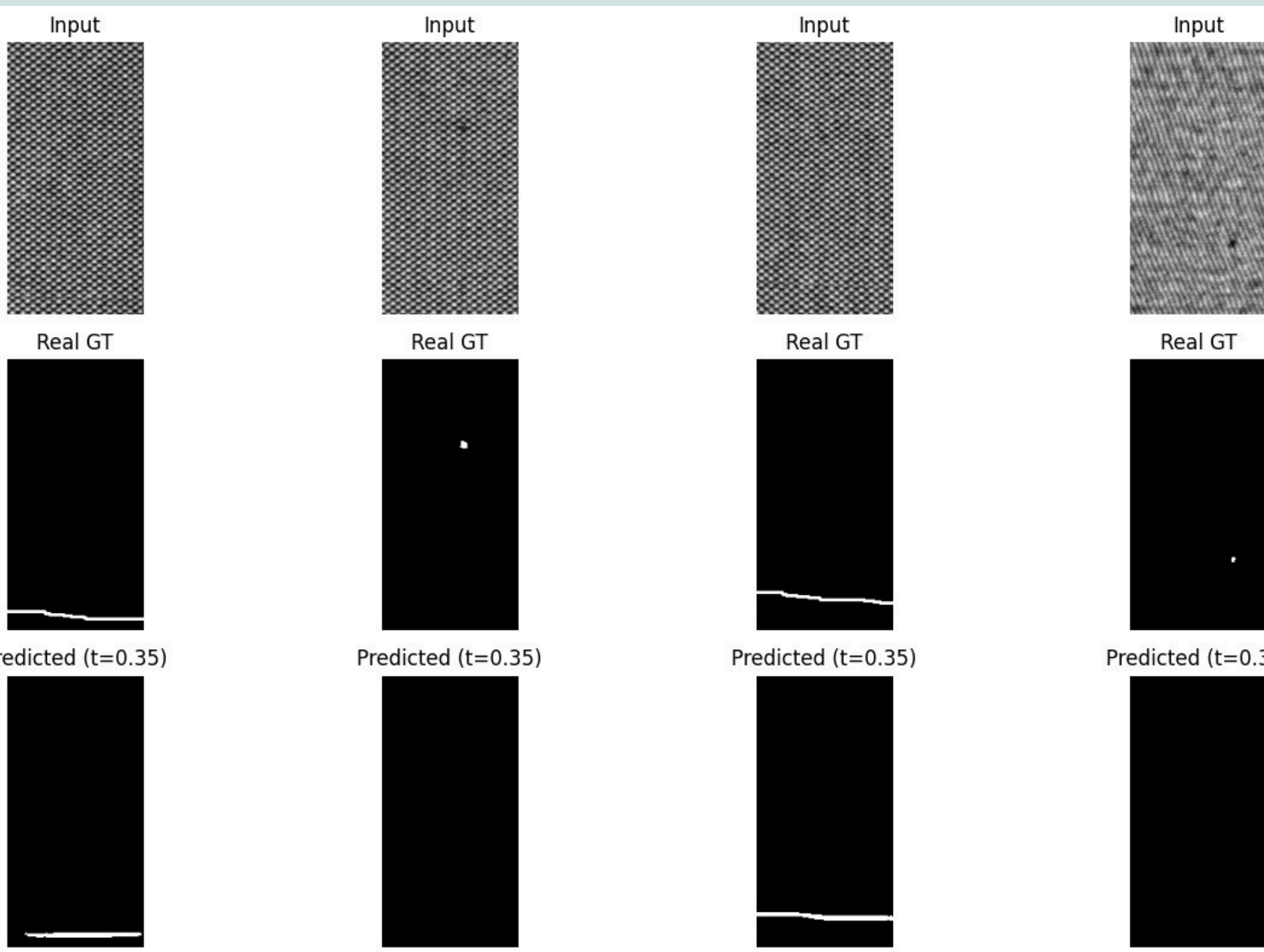


5. RESULTS

Approach	Metric	Value
Classical (RF / XGB)	Accuracy / F1	62% / 0.69
Deep-feature ensemble	CV Accuracy / AUC	89.2% ± 3.9% / 0.94
Autoencoder (anomaly)	Balanced Acc / AUC	80.8% / 0.84
U-Net segmentation	IoU / Dice	0.827 / 0.846



U-Net Segmentation Results



6. DISCUSSION & CONCLUSIONS

The deep-feature-driven CNN ensemble performed the best for overall image classification, achieving an 89.2% cross-validated accuracy and a 0.94 AUC. It significantly outperformed the classical baseline (RF/XGB), which only achieved 62% accuracy, proving that combining deep features with classical descriptors is far more robust at handling the complex, varying textures across the seven different fabric types.

We observed a distinct trade-off between labeling effort and localization precision. The unsupervised Autoencoder provided strong anomaly detection (80.8% balanced accuracy, 0.84 AUC) without ever needing defective images during training. Conversely, the U-Net model provided highly precise spatial localization achieving an IoU of 0.827 and a Dice score of 0.846 but required labor-intensive, pixel-level binary ground truth masks to train.

The primary constraint of this project was the AITEX dataset's extremely small size (only 247 high-resolution images) and its class imbalance (141 defect-free vs. 106 defective images). This required heavy reliance on data augmentation, transfer learning, and patch-based training to prevent overfitting. Future pipelines could address this low-data regime by synthesizing artificial defects to expand the dataset. Additionally, exploring a hybrid model that combines the label-free advantages of an Autoencoder with the precision of a U-Net could yield a more scalable industrial solution.

8. DECLARATION ON USE OF GEN AI

In accordance with Level 3 AI Collaboration guidelines, AI tools (Gemini, ChatGPT]) were used for code drafting, debugging, and text refinement. All code, modeling decisions, results, and references were independently verified by the authors, with full reproducibility details provided in the repository. No AI tools were used for data fabrication or label recovery.